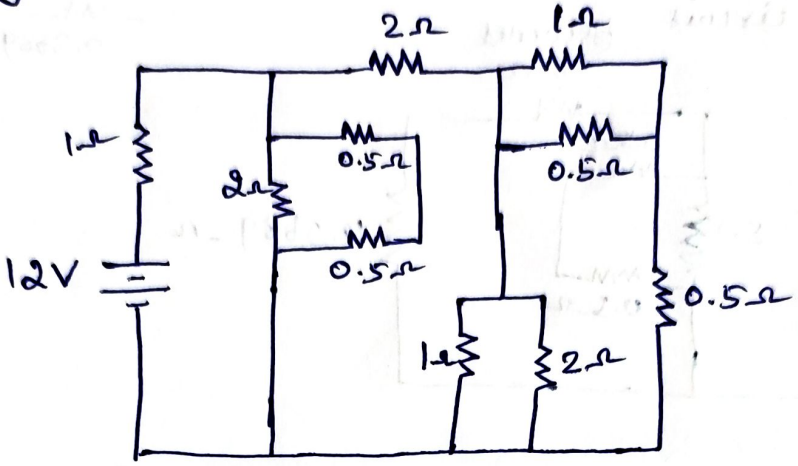
1. a] Find the equivalent resistance & total current to the given circuit.

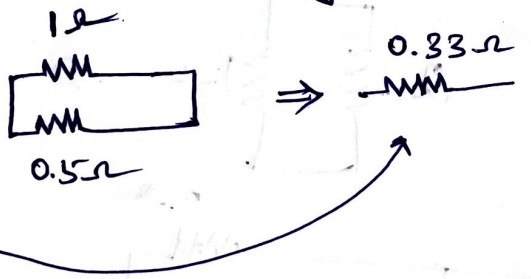


Step 1

We have start calculating the resistance from the opposite end of the Battery.

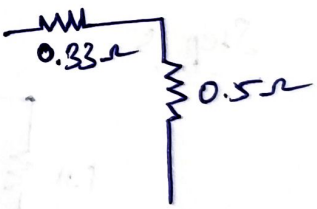
1 & 0.5 ||el

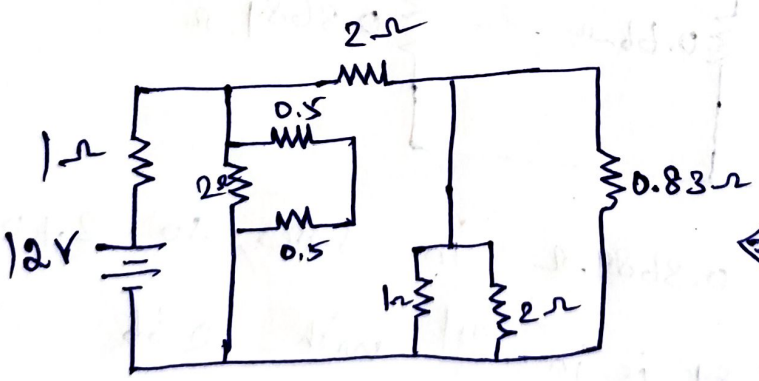
$\frac{1 \times 0.5}{1+0.5} = 0.33$

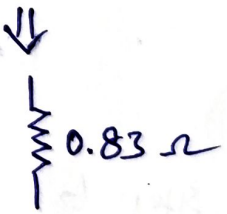


$\Rightarrow$ 0.33Ω

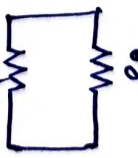
$\Rightarrow$ 0.33 & 0.5 are in series

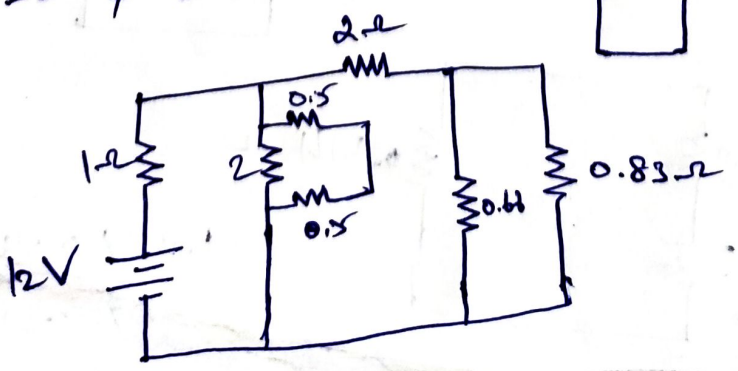




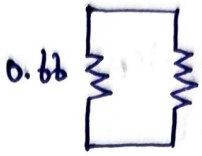


Step 2:

1Ω & 2Ω ||el $\Rightarrow$  $\Rightarrow \frac{1 \times 2}{1+2} = \frac{2}{3} = 0.66\Omega$

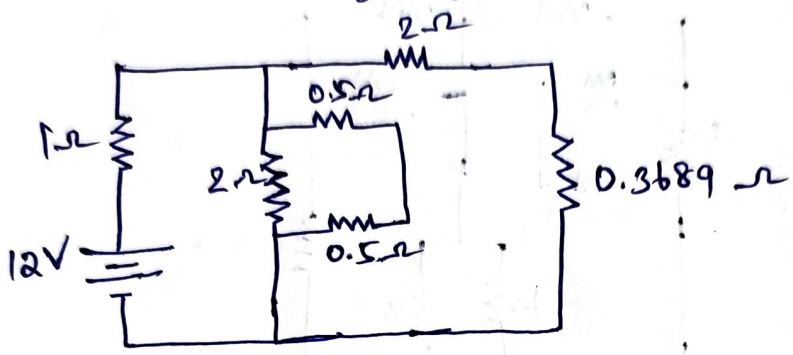


Step 3 — 0.66 Ω & 0.83 in Parallel



$$\Rightarrow \frac{0.66 \times 0.83}{0.66 + 0.83} = \frac{0.553}{1.50} = 0.368$$

Now the circuit becomes

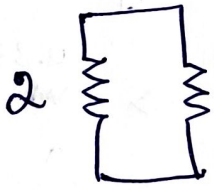


0.3689 Ω

Step 4

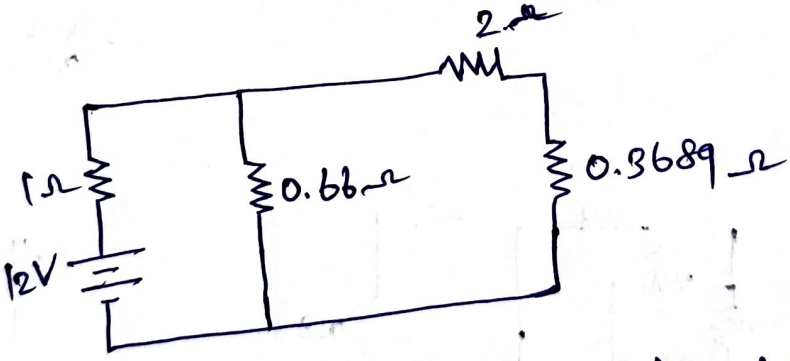
0.5 & 0.5 are in series, so it becomes 0.5 + 0.5 = 1Ω

after that 1 Ω is Parallel with 2 Ω



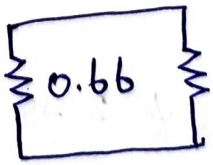
$$\Rightarrow \frac{2 \times 1}{2+1} = \frac{2}{3} = 0.66\ \Omega$$

Step 5



Step 6 — 2 & 0.3689 Ω in series, so 2 + 0.3689 = 2.3689

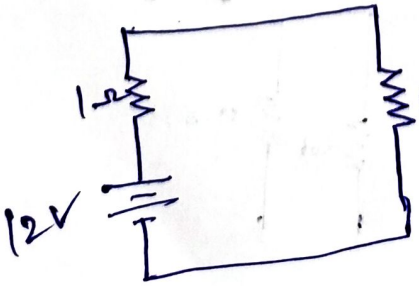
2.3689 is in ||el with 0.66.



$$= \frac{0.66 \times 2.3689}{0.66 + 2.3689}$$

$$= 0.516185$$

Step 6:

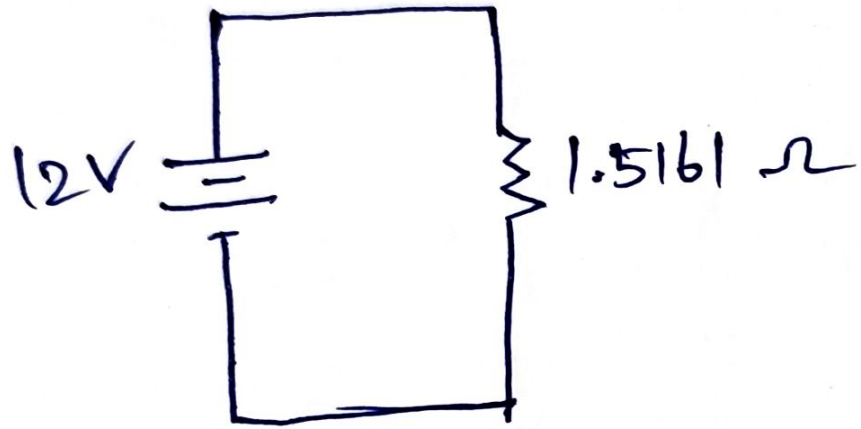


1 Ω & 0.5161 Series

$= 1.5161\ \Omega$

Step 7

$R_{eq} = 1.5161\ \Omega$



$$I = \frac{V}{R_{eq}} = \frac{12}{1.5161}$$

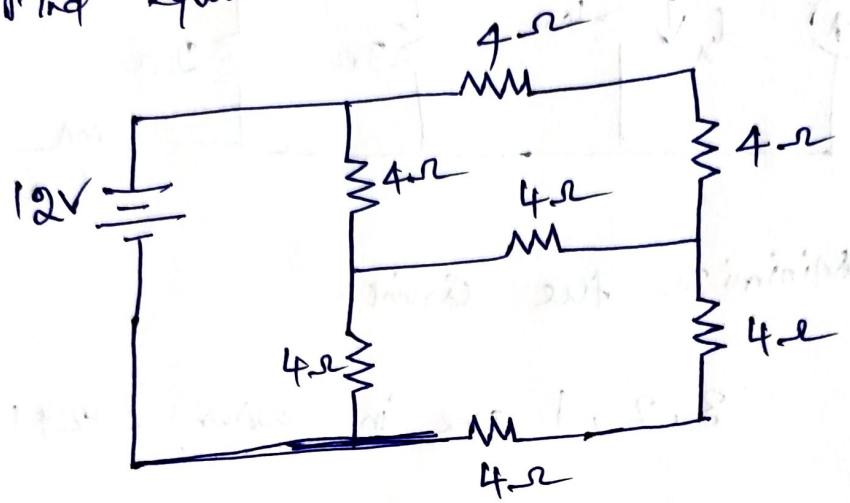
$$= 7.9150\ A$$

Answer:

Total resistance $R_{eq} = 1.5161\ \Omega$

Total current $I = 7.9150\ A$

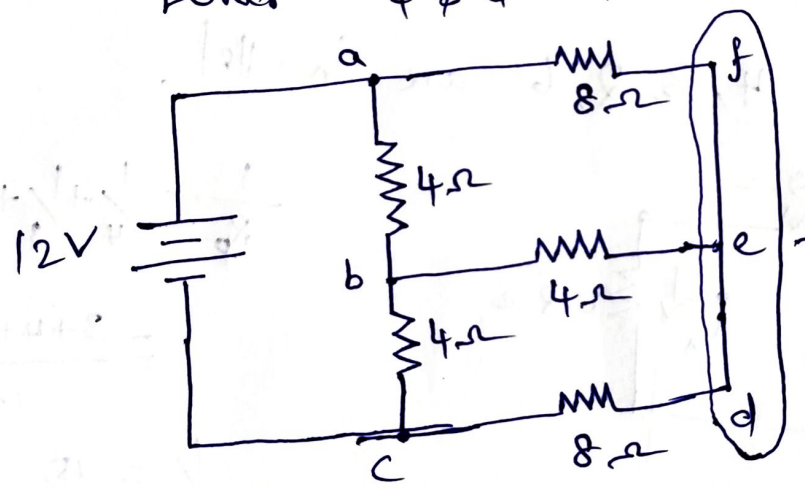
3. a] Find equivalent resistance & total current.



Step 1 Start from the last end.

upper 4 & 4 in series ⇒ 8Ω

Lower 4 & 4 in series ⇒ 8Ω



(d, e, f) → This ~~last~~ not point having any resistance.

So lets take only one point as d.

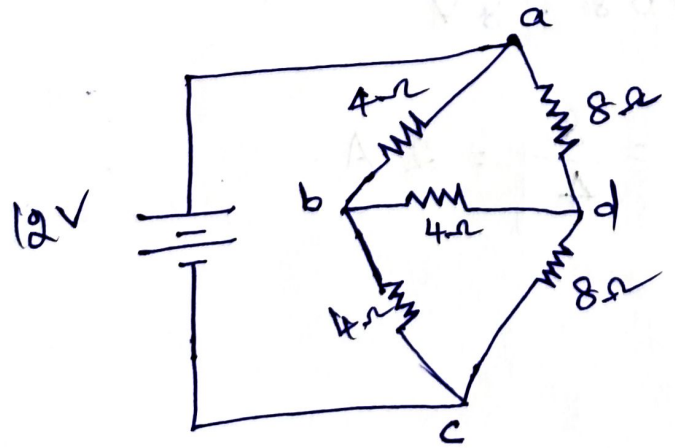
From the above ckt

∴ a - d = 8Ω

b - d = 4Ω

c → d = 8Ω

Step 2 ⇓ Transform



It looks like Balanced Wheatstone Bridge.

(B → d = current zero)

So, R_{eq} = 4 + 4 ⇒ Series = 8

= 8 + 8 ⇒ Series = 16

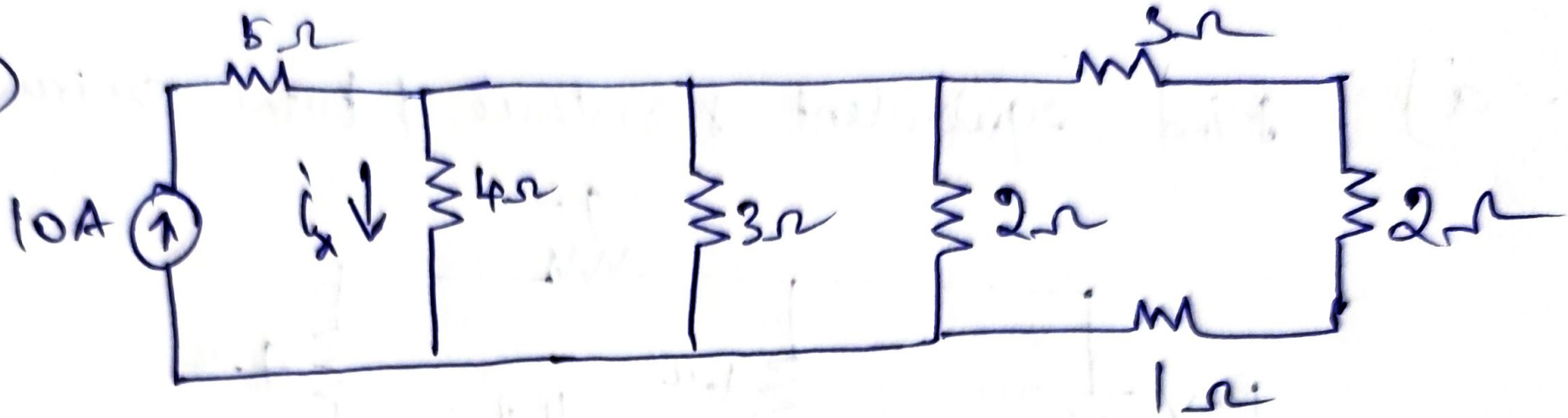
Then 8 || 16

$= \frac{8 \times 16}{8+16} = \frac{128}{24} = 5.33\Omega$

$R_{eq} = 5.33\Omega$

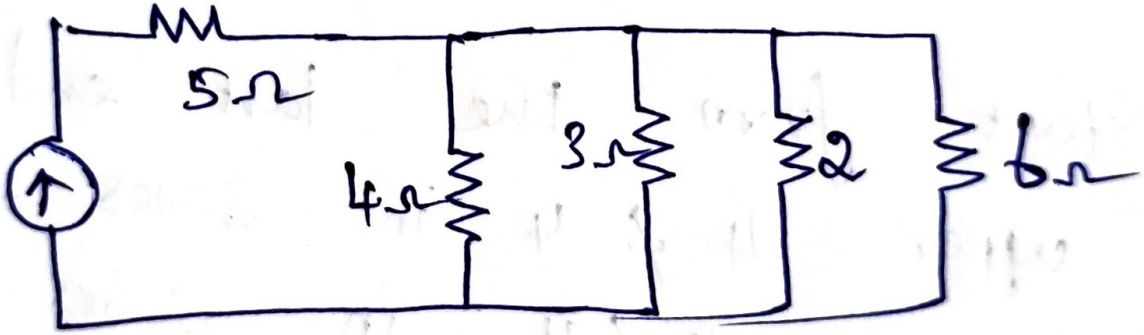
$I = \frac{V}{R_{eq}} = \frac{12}{5.33} = 2.25A$

4. a)



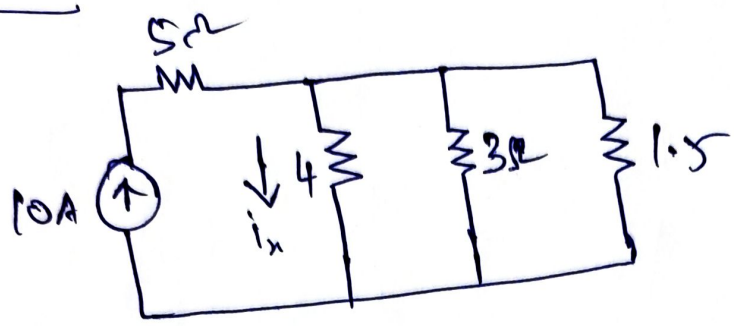
Step 1 Minimize the circuit.

3, 2, 1 are in series $3+2+1 = 6\Omega$



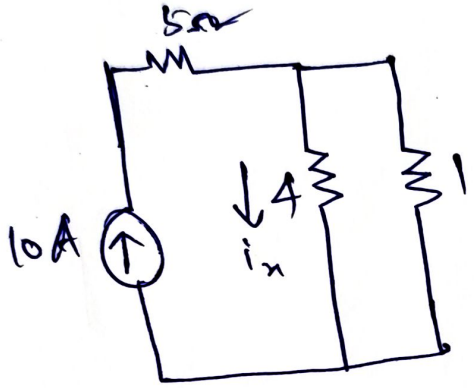
Step 2

$6 \parallel 2$ (Rep) $= \frac{6 \times 2}{6+2} = \frac{12}{8} = 1.5\,\Omega$



Step 3

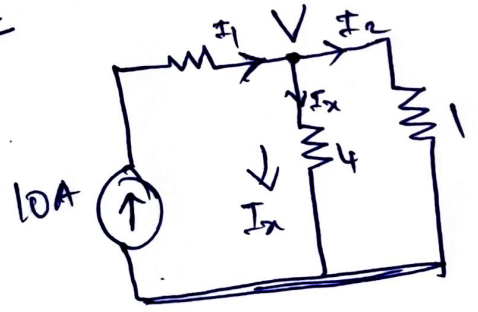
$3 \parallel 1.5 = \frac{3 \times 1.5}{3+1.5} = \frac{4.5}{4.5} = 1\,\Omega$



Step 4

Either apply KCL (or) KVL (or) current division

Method: KVL



KCL

$$I_1 = I_x + I_2$$

$$10 = \frac{V-0}{4} + \frac{V-0}{1}$$

$$10 = \frac{V}{4} + \frac{V}{1}$$

$$10 = \frac{5V}{4}$$

$$V = \frac{40}{5} = 8V$$

$$I_x = \frac{V-0}{4} \Rightarrow \frac{8}{4} = 2A$$

Method 2 current divisibn.

$$i_x = I_{total} \times \frac{R_{opposite\ branch}}{R_{branch} + R_{opposite}}$$

$$i_x = 10 \times \left(\frac{1}{4+1}\right)$$

$$i_x = 10 \times \frac{1}{5}$$

$$i_x = 2\ A.$$